Lazy Distillation for Long-Horizon Self-Evolving Agents

Breaking Confirmation-Bias Feedback Loops while Preserving Workflow Patterns

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Abstract

Self-evolving agents repeatedly collect multi-step trajectories, construct pseudo-targets from their own policy and environment feedback, and internalize those targets through parameter updates. This creates a feedback loop that is absent in one-shot distillation: an early error changes the policy, the changed policy visits different states, and those states supply the next training set. We isolate two coupled long-horizon failures: *confirmation bias*, in which a wrong policy action becomes its own pseudo-label, and *pattern loss*, in which sequential non-IID updates overwrite previously acquired minority workflows. We propose LazyDistill, which freezes the trajectory collector for a multi-wave window and performs one pooled update. In an 8-step, 12-cycle controlled agent benchmark, LazyDistill obtains $74.0\% \pm 17.8\%$ episode success while vanilla eager self-distillation collapses to 0%. Against confidence filtering, Mean Teacher, and replay, LazyDistill yields the lowest pattern loss ($3.6\% \pm 3.1\%$). At 16 steps and 24 cycles, Lazy retains $79.6\% \pm 0.8\%$ success and $6.4\% \pm 4.9\%$ pattern loss, while eager self-distillation again collapses to zero success. These results are preliminary: the benchmark is procedural and the policy is lightweight, so external validation on language-agent environments remains necessary.

1 Motivation

Language agents can improve from their own interaction traces without requiring a stronger teacher for every action. STaR filters self-generated rationales [Zelikman et al., 2022]; ReST reuses offline model samples [Gulcehre et al., 2023]; ReST-meets-ReAct extends this idea to multi-step agents [Aksitov et al., 2023]; and Re-ReST repairs weak trajectories using environment feedback [Dou et al., 2024]. However, most headline evaluations use one or a few improvement rounds. A genuinely self-evolving agent runs the following loop many times:

$$\pi_k \longrightarrow \mathcal{D}_k(\pi_k) \longrightarrow \widehat{\mathcal{D}}_k \longrightarrow \pi_{k+1}, \tag{1}$$

where the current policy π_k determines the trajectory distribution $\mathcal{D}_k$, and therefore determines its own future supervision.

Our central premise is deliberately narrow: **confirmation bias and pattern loss become a coupled agent failure under long-horizon, repeated self-evolution**. In one-shot distillation there is no recursive $\pi_{k+1} \to \mathcal{D}_{k+1}$ contamination. In long-horizon interaction, a single wrong action can move the agent into a different state subtree; immediate learning can then make that subtree more likely in every later round.

Closest concurrent evidence. Chen et al. [2026] independently report progressive capability collapse under multi-iteration experience internalization and find stable off-policy context distillation preferable to on-policy internalization. Shenfeld et al. [2026] show that demonstration-conditioned self-distillation can reduce continual forgetting. We therefore do not claim that iterative self-training failure is new in the broad sense. Our target contribution is a controlled causal decomposition of *update timing*, together with separate probes for error feedback and pattern erasure.

Direct self-evolving-agent baselines. SDAR combines GRPO with on-policy self-distillation from a privileged-context teacher and applies a detached, token-level positive-gap gate [Lu et al., 2026]. SEED distills hindsight skills from completed trajectories [Wu et al., 2026], while OPID uses hierarchical episode- and step-level skills [Yang et al., 2026]. These methods make agent self-distillation substantially more relevant than generic semi-supervised baselines. They nevertheless update on the current policy's experience; our question is orthogonal: what happens when such experience internalization is repeated over a long, non-IID evolution horizon?

2 Failure Setting

An episode is a partially observed workflow with horizon H. At state s_t, the policy selects one of eight tools. The hidden workflow program defines the oracle action $a_t^\star$. A weak hindsight verifier and the policy distribution form a pseudo-label:

$$q_k(a \mid s) = (1-\lambda)\,\mathrm{Sharpen}(\pi_k(a \mid s)) + \lambda v(a \mid s), \qquad \hat{a} = \arg\max_a q_k(a \mid s). \tag{2}$$

The verifier is usually correct but deliberately too weak to override an already overconfident wrong policy. This does not force confirmation bias: it creates the condition under which bias can either grow or be corrected.

2.1 Probing confirmation bias

Because the simulator hides an oracle action from the learner, we can diagnose pseudo-targets without changing training. We report:

$$\mathrm{PseudoErr}_k = \Pr[\hat{a}_k \neq a^\star], \tag{3}$$

$$\mathrm{SelfReinforce}_k = \Pr[\arg\max_a \pi_k(a \mid s) \neq a^\star \ \wedge\ \hat{a}_k = \arg\max_a \pi_k(a \mid s)], \tag{4}$$

$$\mathrm{Amplification} = \mathrm{PseudoErr}_K - \mathrm{PseudoErr}_1. \tag{5}$$

We additionally track normalized pseudo-action entropy. Falling entropy together with rising self-reinforced error distinguishes collapse from merely hard data.

2.2 Probing pattern loss

Workflows arrive in non-IID waves. For pattern p, let k_p be its first exposure and $S_{p,k}$ its full-episode success after update k. We define

$$\textsc{PatternLoss}(p) = \max_{k \geq k_p} S_{p,k} - S_{p,K}. \tag{6}$$

This asks whether the agent *previously knew* a workflow and lost it after later self-evolution. Frozen has zero pattern loss mechanically because it never updates; it is an ability anchor, not a continual-learning solution.

3 Lazy Distillation

LazyDistill separates collection time from internalization time. For a window of W non-IID waves, it holds the collector fixed at $\bar{\pi}_j$:

$$\mathcal{B}_j = \bigcup_{w=1}^{W} \widehat{\mathcal{D}}_{j,w}(\bar{\pi}_j). \tag{7}$$

Only after all waves arrive does it update on the pooled buffer:

$$\pi_{j+1} = \arg\min_\pi \sum_{x \in \mathcal{B}_j} \ell(\pi(\cdot \mid s_x), \hat{a}_x)\,. \tag{8}$$

Why it targets confirmation bias. Within a window, an early wrong update cannot change later trajectories because there is no within-window update. The loop gain of error→parameter→data is therefore reduced from every wave to once per window.

Why it targets pattern loss. Common and rare waves are co-present in the same optimization event. Their gradients compete jointly instead of the final dominant wave overwriting all earlier waves. This is different from replay: replay repairs a sequential update after it occurs and may repeatedly replay corrupted pseudo-labels.

Table 1: Eight-step, 12-cycle results over five seeds (mean ± s.d.). Arrows indicate the preferred direction.

Method	Success ↑	Pseudo error ↓	Self-reinforce ↓	Pattern loss ↓
Frozen	35.8 ± 5.9	46.4 ± 7.1	44.9 ± 7.2	0.0 ± 0.0
Eager	0.0 ± 0.0	94.4 ± 1.2	94.3 ± 1.2	39.6 ± 9.7
Confident eager	54.0 ± 30.2	56.8 ± 17.2	56.5 ± 17.2	26.8 ± 14.4
Mean Teacher	62.8 ± 12.9	49.0 ± 8.1	48.2 ± 8.2	24.3 ± 9.3
Eager + replay	56.6 ± 9.3	60.4 ± 8.1	60.2 ± 8.1	10.5 ± 2.6
Lazy (timing)	$\mathbf{74.0 \pm 17.8}$	$\mathbf{43.2 \pm 12.5}$	$\mathbf{43.0 \pm 12.5}$	$\mathbf{3.6 \pm 3.1}$

4 Experimental Setup

4.1 Dataset and environment

We generate 24 procedural workflow patterns organized into 12 confusable common/rare pairs. Pair members have highly similar observation cues but disagree on half of their hidden action programs. Each observation includes the workflow cue, current progress, previous tool call, and success/failure feedback. Later cycles oversample common patterns, creating a realistic dominant-trajectory stream without deleting minority patterns.

The main setting uses $H = 8$, three attempts per stage, six waves, 12 evolution cycles, and five seeds. The scale setting uses $H = 16$, 24 evolution cycles, and three seeds. Every method starts from the same warm-start policy and sees the same schedule. Results report mean and sample standard deviation.

4.2 Comparison methods

- **Frozen**: no parameter updates.
- **Eager self-distillation**: update after every wave.
- **Confidence-filtered eager**: retain pseudo-labels above a fixed confidence threshold, following pseudo-label selection defenses [Radhakrishnan et al., 2023, Rodemann et al., 2023].
- **Mean Teacher**: use an exponential moving-average teacher [Tarvainen and Valpola, 2017].
- **Eager + replay**: mix each current wave with historical trajectories, following continual-learning replay [Yoo et al., 2024] and growing-batch ReST [Gulcehre et al., 2023].
- **SDAR-style gated eager**: adapt SDAR's detached positive-gap gate [Lu et al., 2026] to our action-level policy. For pseudo-target $\hat{a}$, $g = \sigma(\beta[\log q(\hat{a} \mid s) - \log \pi(\hat{a} \mid s)])$ is detached and multiplies cross-entropy. This isolates SDAR's gating mechanism but is *not* an exact reproduction of its token-level GRPO system.
- **Lazy**: freeze collection for six waves and then update on all collected pseudo-labels.

These implementations isolate the papers' relevant mechanisms; they are not claimed as exact reproductions on the original datasets.

5 Preliminary Results

5.1 Does the long self-evolution failure exist?

Table 1 first establishes the phenomenon. Eager self-distillation collapses from a nonzero frozen policy to zero full-episode success. Its pseudo-label error reaches 94.4% on average, and nearly every wrong greedy action becomes its own target. Its pseudo-action entropy also falls, consistent with mode collapse rather than uniformly hard tasks.

5.2 Probing whether lazy learning addresses both failures

The comparison separates three partial defenses. Confidence filtering is high-variance because the evolving policy can become confidently wrong. Mean Teacher stabilizes targets but still performs sequential non-IID updates. Replay sharply reduces pattern loss but retains high pseudo-label error because old incorrect targets are replayed. Lazy timing is the only updating method in Table 1 with both the lowest error feedback and near-zero pattern loss. A separate probe found that naively adding a confidence gate to Lazy can hurt at the longest horizon because the evolving policy becomes confidently wrong. This negative result strengthens the timing interpretation: confidence selection is not a substitute for breaking the recursive update loop.

Table 2: Sixteen-step, 24-cycle long-horizon scale experiment.

Method	Success ↑	Pseudo error ↓	Self-reinforce ↓	Pattern loss ↓
Frozen	52.9 ± 3.5	24.0 ± 3.5	21.9 ± 3.5	0.0 ± 0.0
Eager	0.0 ± 0.0	95.3 ± 2.5	95.2 ± 2.6	19.6 ± 8.6
Confident eager	0.0 ± 0.0	93.6 ± 1.7	93.5 ± 1.7	12.2 ± 3.9
Mean Teacher	79.1 ± 9.6	30.3 ± 9.1	29.8 ± 9.1	11.3 ± 4.4
Eager + replay	24.3 ± 4.0	69.4 ± 1.6	69.3 ± 1.6	10.9 ± 7.6
SDAR-style gate	3.3 ± 3.0	88.3 ± 4.0	88.0 ± 4.1	41.0 ± 15.4
Lazy	$\mathbf{79.6 \pm 0.8}$	$\mathbf{29.9 \pm 2.9}$	$\mathbf{29.8 \pm 2.9}$	$\mathbf{6.4 \pm 4.9}$

5.3 Scale with horizon and evolution depth

The decisive scale experiment increases both axes that generate the recursive failure: horizon from 8 to 16 and evolution cycles from 12 to 24. The final three-seed values are shown in Table 2.

5.4 Why SDAR-style gating does not replace lazy timing

The SDAR-style adaptation reaches only $3.3\% \pm 3.0\%$ success and accumulates $88.3\% \pm 4.0\%$ pseudo-label error. This negative result should not be read as a leaderboard claim against full SDAR: our controlled policy has neither language tokens nor GRPO rewards nor a privileged skill-context teacher. Instead, it is a causal probe of the transferable mechanism. A positive-gap gate controls *how strongly* the learner fits each current target, but eager updates still change the collector after every wave. Once a wrong pattern becomes confident, its target can retain substantial weight, affect later state visitation, and overwrite earlier patterns.

Lazy changes the missing axis: *when* learning is allowed to affect data collection. Freezing the collector blocks within-window error→data feedback, and pooling all waves makes minority and majority patterns co-present. Thus SDAR-style gating and Lazy are conceptually orthogonal; a full Lazy-SDAR experiment on ALFWorld or WebShop is an important next step.

6 Limitations and Next Steps

This is a procedural stress test with a lightweight policy and a simulator oracle. It is designed for causal probing, not external validity. The hyperparameter regime was calibrated on one seed before multi-seed evaluation. The replay baseline receives additional historical examples, so it is data-reuse matched rather than optimizer-step matched. Our next experiment should preserve the same frozen-window intervention in ALFWorld, WebShop, or a comparable language-agent environment. The SDAR row is a mechanism adaptation rather than an exact reproduction; a fair external comparison requires the authors' GRPO stack, language-model teacher, and original ALFWorld/SearchQA/WebShop protocols.

7 Conclusion

Long-horizon self-evolution turns ordinary pseudo-label noise into a dynamical system: the learner changes the data that will supervise its next version. Preliminary results show that eager updates can simultaneously amplify wrong actions and erase old workflows. Lazy collection attacks the causal link responsible for the first failure and pooled optimization attacks the sequential overwrite responsible for the second. Naive confidence gating is not sufficient when confidence itself is corrupted.

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